



Introduction to Cybersecurity

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Lecture 3

Lecture Rules



Cyberattacks and Vulnerabilities

Part I

Lecture Outline

- **Importance of Cybersecurity.**
- **Cyberattack Vs. Vulnerability.**
- **Types of Cyberattacks:**
 - **Denial of Service (DoS) attacks.**
 - **Distributed Denial of Service (DDoS) attacks.**
 - **DoS Vs. DDoS attacks.**

Importance of Cybersecurity

1. According to the latest predictions from the **Gartner research organization**, the **total global** spending in the field of **cybersecurity** is expected to cross the **US\$124 billion** mark in **2019** with a huge growth of **over 8.7%**.
2. The growth rate of security spending in **2018** as compared to that of **2017** was estimated at **about 12.4%**.
3. The total spending in **2018** is estimated at about **US\$114 billion**.
4. The total global spending related to **cybersecurity** is **expected** to cross the **US\$1 trillion** mark by **2024**. The cybercrime cost in 2025 is about **US\$ 9 trillion**

Thus, this huge cybersecurity spending will make it one of the **biggest domains** in the **information technology field**.

Importance of Cybersecurity

5. **Many** kinds of **software** and **hardware** tools are being used on the **networks**, **computers**, **storage devices**, **servers**, **applications**, and **vulnerable codes** which can be attacked by **hackers**.
6. **The security** of **network** has got much spread in the present days due to the **strong security** features and **capabilities** of **powerful firewalls**.
7. **The applications** and **software programs** are the **most vulnerable** areas for the **hackers** to attack on.

Cyberattack Vs. Vulnerability

- **Cyberattack** refers to getting access to the legitimate systems – **servers, computers, network, or software programs** – **illegally** and establishing control over the systems for carrying out **malicious activities**.
- These **activities** such as **information stealth, data damage, system damage**, and **interruption** in **smooth operations** of the **legitimate networks** and **systems**.

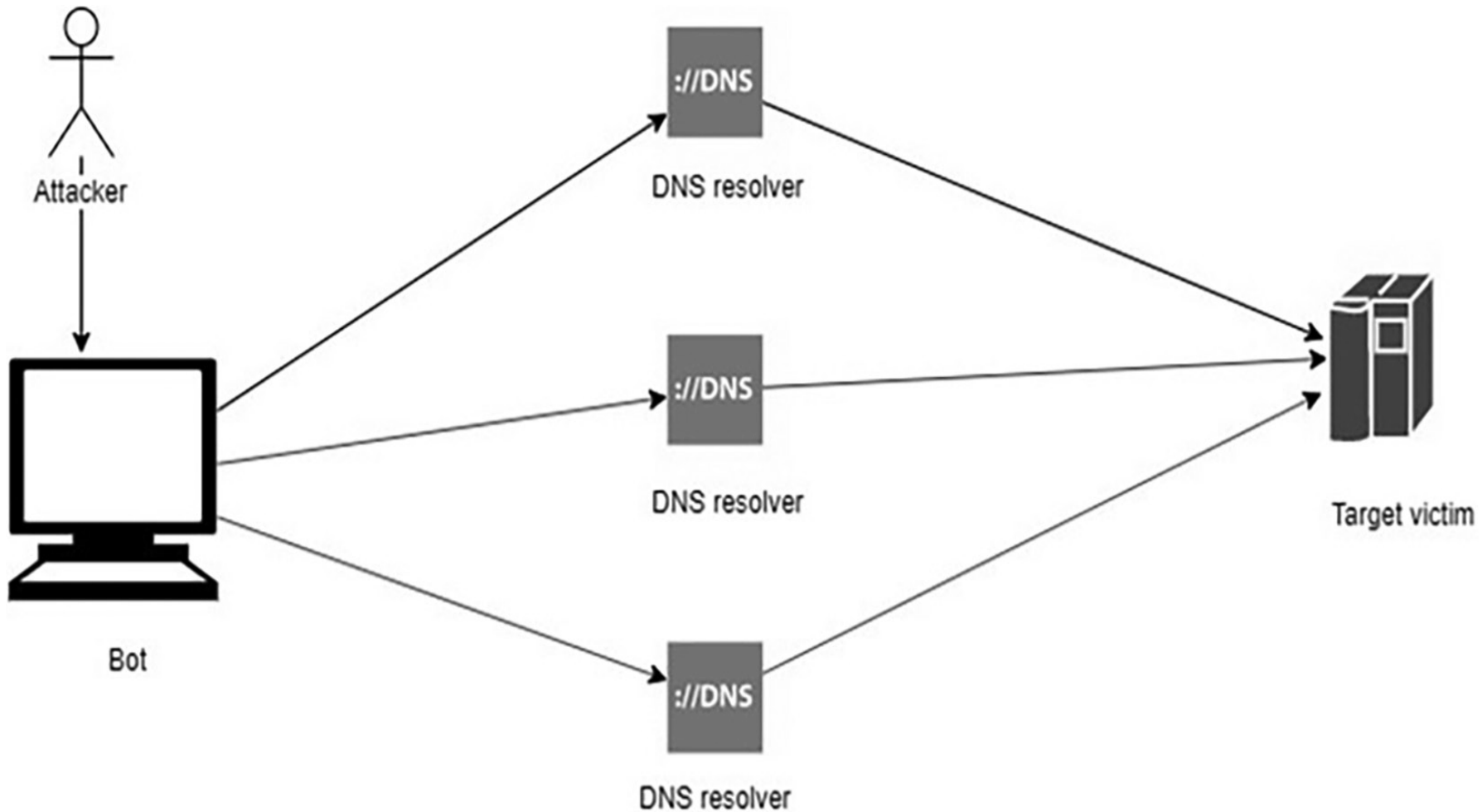
Cyberattack Vs. Vulnerability

- A **vulnerability** refers to **any weakness** in an **organization's digital infrastructure** that could be **exploited by hackers**.
- These **weaknesses** could be **anything** from **outdated software programs** with **known security flaws** to **unsecured networks**.

1. Denial of Service (DoS)

- The hacker actively **exploits** the **server vulnerability** and sends the **huge amount** of **automated requests** and **messages** to that **particular** server to **respond**. So, The server gets **choked** and **stops** working **normal**.
- In certain cases, the **service stops** working **due to overloaded** server.
- Day-to-day cyberattacks on networks and computers to either **steal valuable** data or **damage** the **continuity** of **online services** on the **Internet**.

1. Denial of Service (DoS)



1. Denial of Service (DoS)

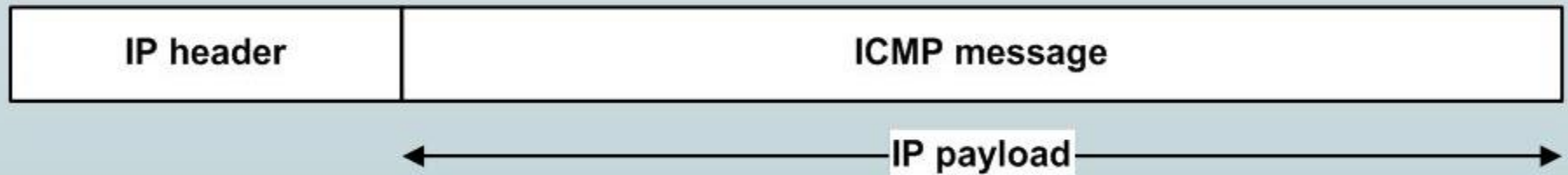
- The major **symptoms** of being the **victim** of DoS attacks **(for a legitimate user)** include the following:
 - **Inability** in **accessing** a **website**.
 - **Delay** in **accessing online** service.
 - **Huge delays** in **file opening** on the **websites**.
 - **Increased** volume of **spam emails**.

1. Denial of Service (DoS)

- Main types of **DoS attacks** include the following:
 - **ICMP** (Internet Control Message Protocol) **flooding**.
 - **Network** attack or **buffer overflow attack**.
 - **Ping of death (PoD)** attack.
 - **SYN flood** attack on **TCP handshake protocol**.
 - **DNS** (Domain Name System) **server attack**.
 - **HTTP** (Hypertext Transfer Protocol) **server attack**.

ICMP

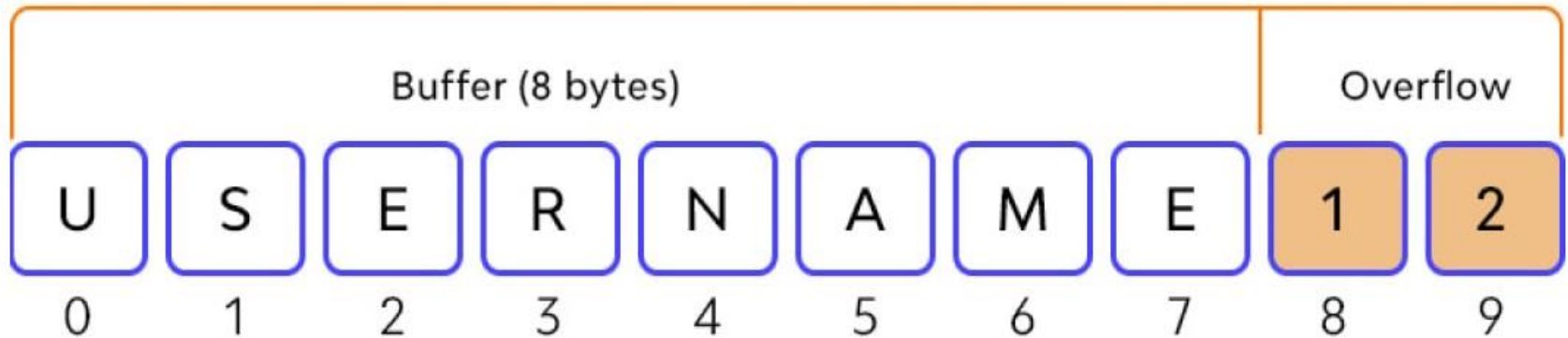
- The **Internet Control Message Protocol (ICMP)** is a helper protocol that supports IP with facility for
 - Error reporting
 - Simple queries
- ICMP messages are encapsulated as IP datagrams:



- **ICMP flood attack** is a common DoS attack where **hackers** try to **crush** a **server** or **network** device with **ICMP requests (pings)**. ICMP requests are used to determine the **integrity** of **device and connection**.

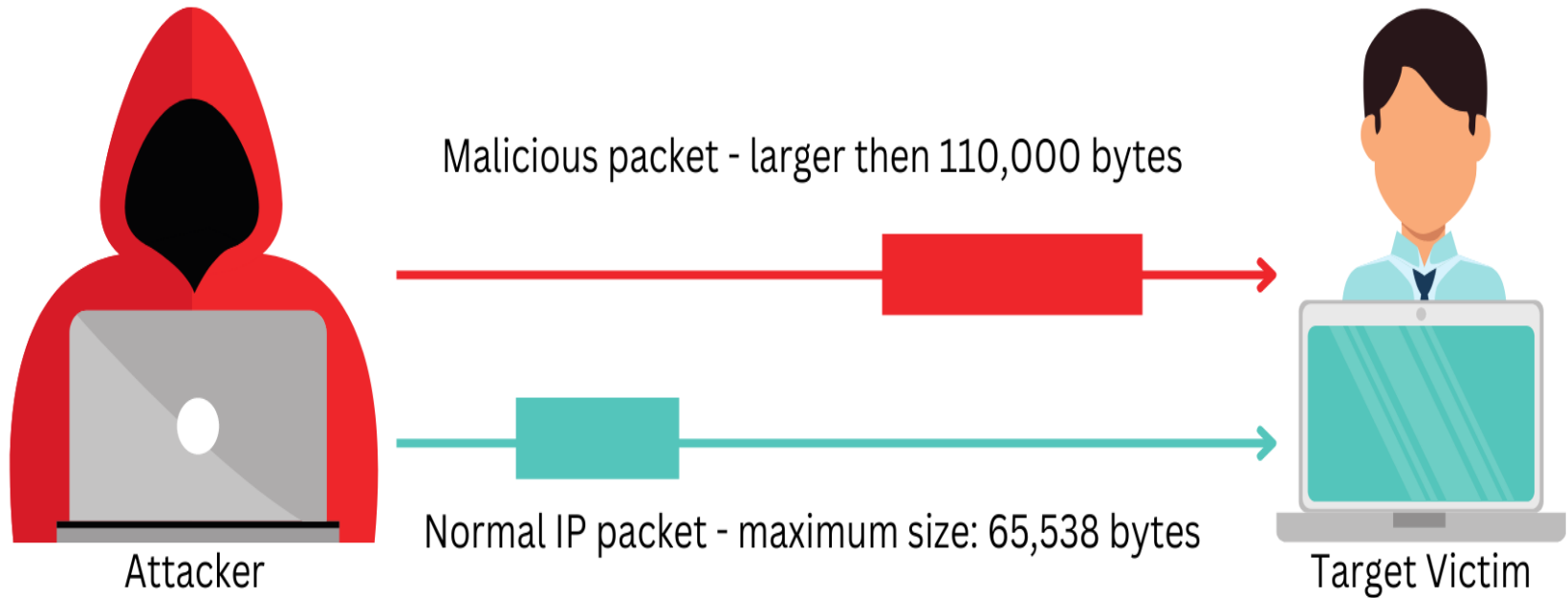
Buffer overflow attack

Buffer overflow example



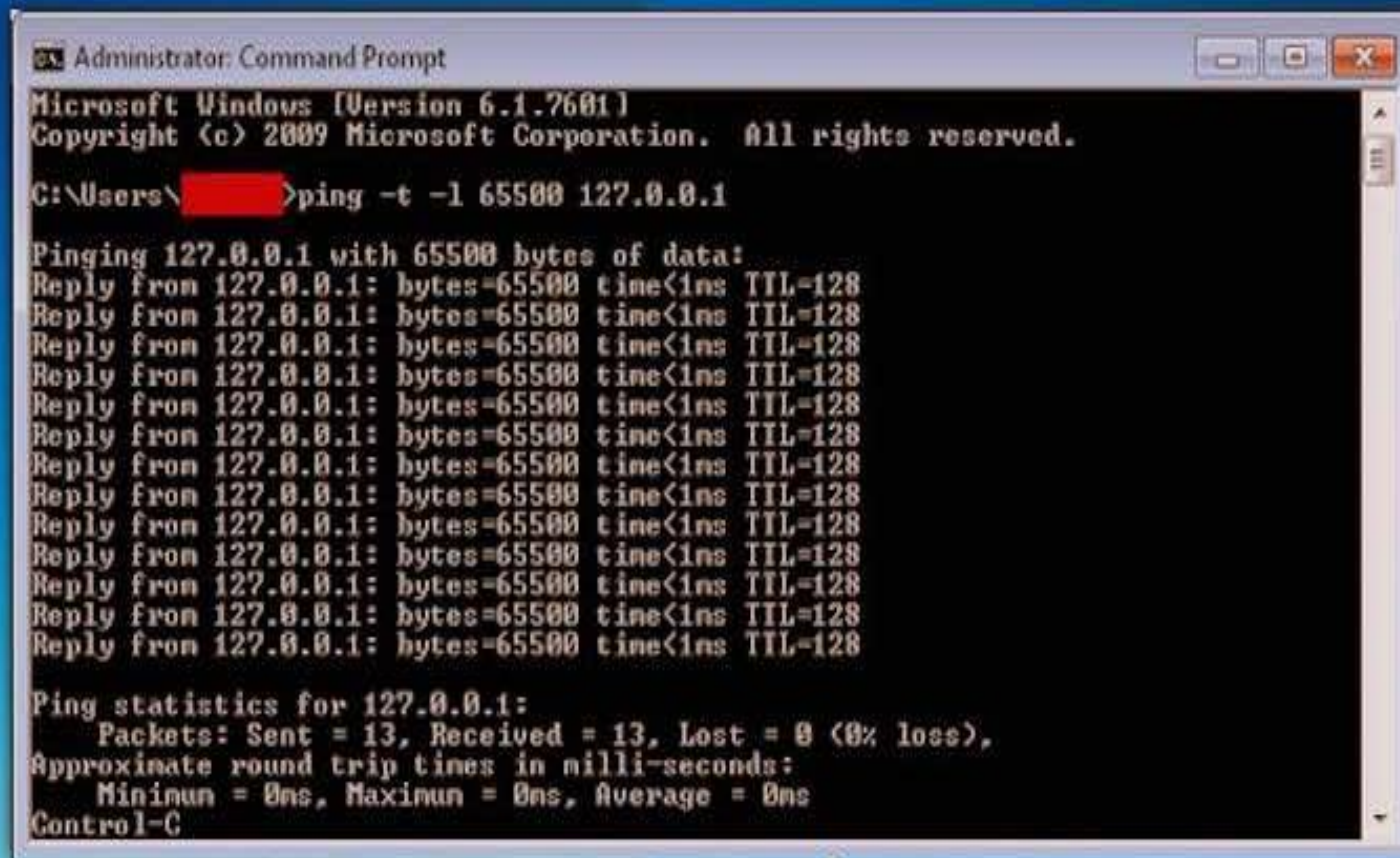
- The input **"USERNAME12"** is longer than the **buffer** can **hold**. The extra **"12"** overflows into **memory**, so **corrupting** data or allowing **malicious code** to be **injected**.
- **Buffer overflow attack** is a **software coding error** or **vulnerability** that can be **exploited** by **hackers** to **gain unauthorized access** to **legitimate systems**.

Ping of Death Attack



- The **ping of death (PoD) attack** is a **DoS attack** that sends out **abnormally large packets** to **disable, crash, or unstable** a **web server (Offline mode)**.

Ping of Death Attack



The screenshot shows a Windows Command Prompt window titled "Administrator: Command Prompt". The window displays the output of a ping command: `ping -t -l 65500 127.0.0.1`. The output shows 13 successful replies from 127.0.0.1, each with 65500 bytes of data, a time of less than 1ms, and a TTL of 128. The ping statistics show 13 packets sent, 13 received, and 0 lost (0% loss). The approximate round trip times are all 0ms. The window is set against a blue background with a "Recycle Bin" icon visible in the top left corner.

```
Administrator: Command Prompt
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

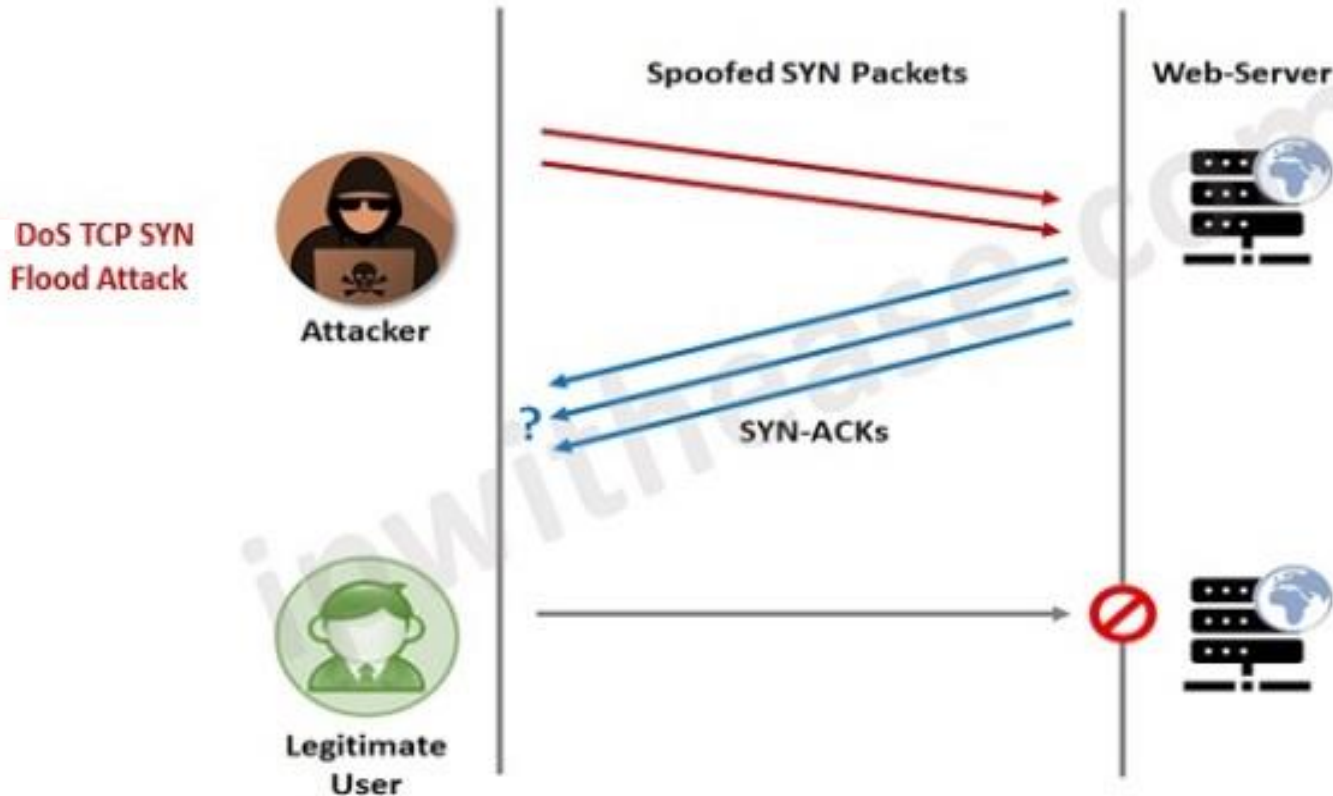
C:\Users\[redacted]>ping -t -l 65500 127.0.0.1

Pinging 127.0.0.1 with 65500 bytes of data:
Reply from 127.0.0.1: bytes=65500 time<1ms TTL=128
Reply from 127.0.0.1: bytes=65500 time<1ms TTL=128
Reply from 127.0.0.1: bytes=65500 time<1ms TTL=128
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Ping statistics for 127.0.0.1:
    Packets: Sent = 13, Received = 13, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
Control-C
```

PING OF DEATH ATTACK

TCP SYN Flood Attack

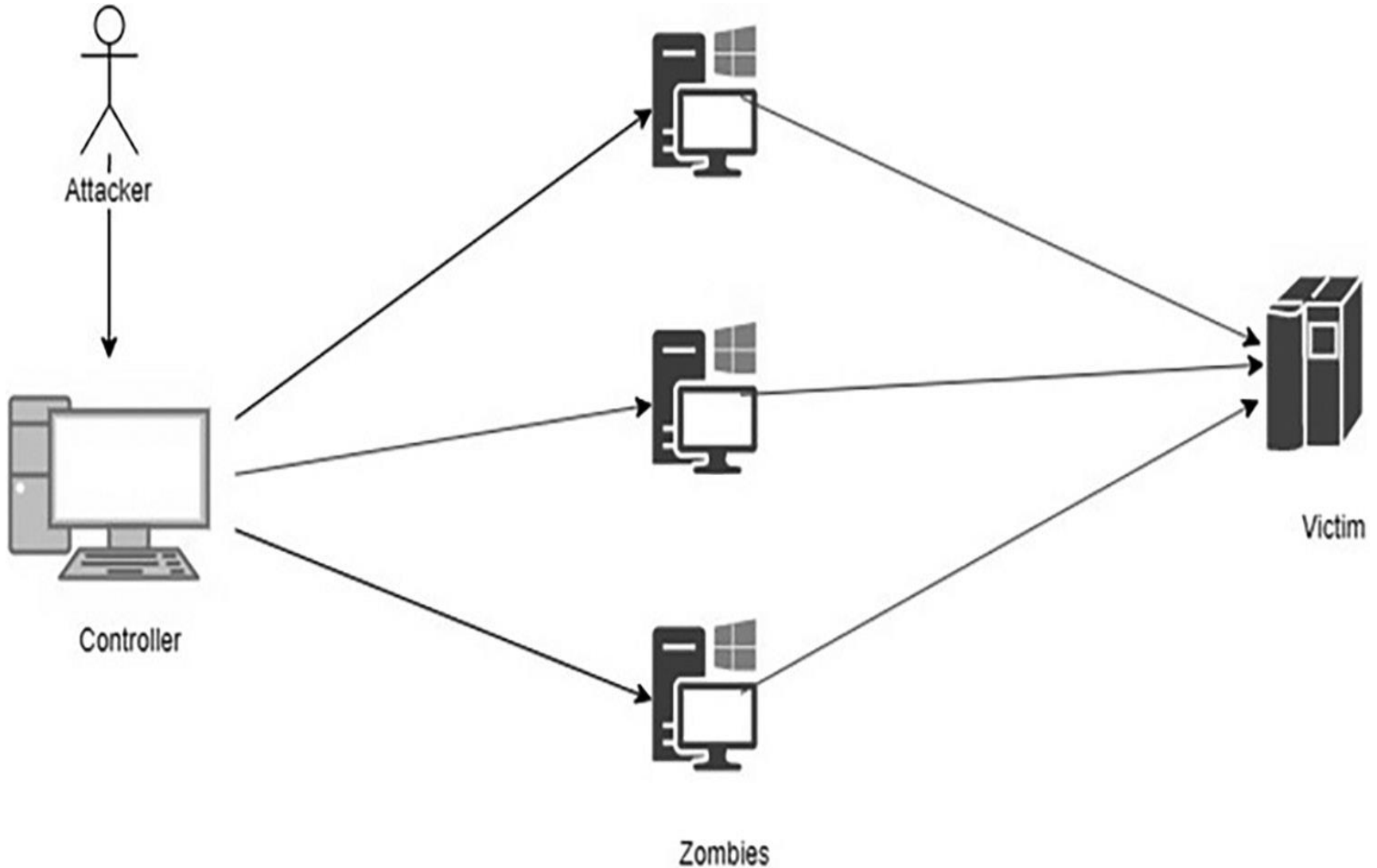


- **eShopper.com** becomes unavailable to legitimate customers
- A **SYN flood** exploits a **vulnerability** in the **TCP/IP handshake** in an attempt to **disable** or **crash** a **web service**.

2. Distributed Denial of Service (DDoS)

- In the DDoS attack, the **hacker** uses **controller** machine to hack **multiple vulnerable machines** around the globe.
- Then, all those **infected machines** will be **controlled** by the **software program** running on the **attacker's server**.
- That **controller** sends **instructions** to the **infected machine** on which the **hacker** has already established the **control** to send **automated requests** (traffic) to the targeted **victim** machine.
- Then, all those “**zombies**” start **attacking** the **target** from **multiple IP sources** to bring the **server** to almost a **halt (down)**.

2. Distributed Denial of Service (DDoS)



2. DDoS Attacks

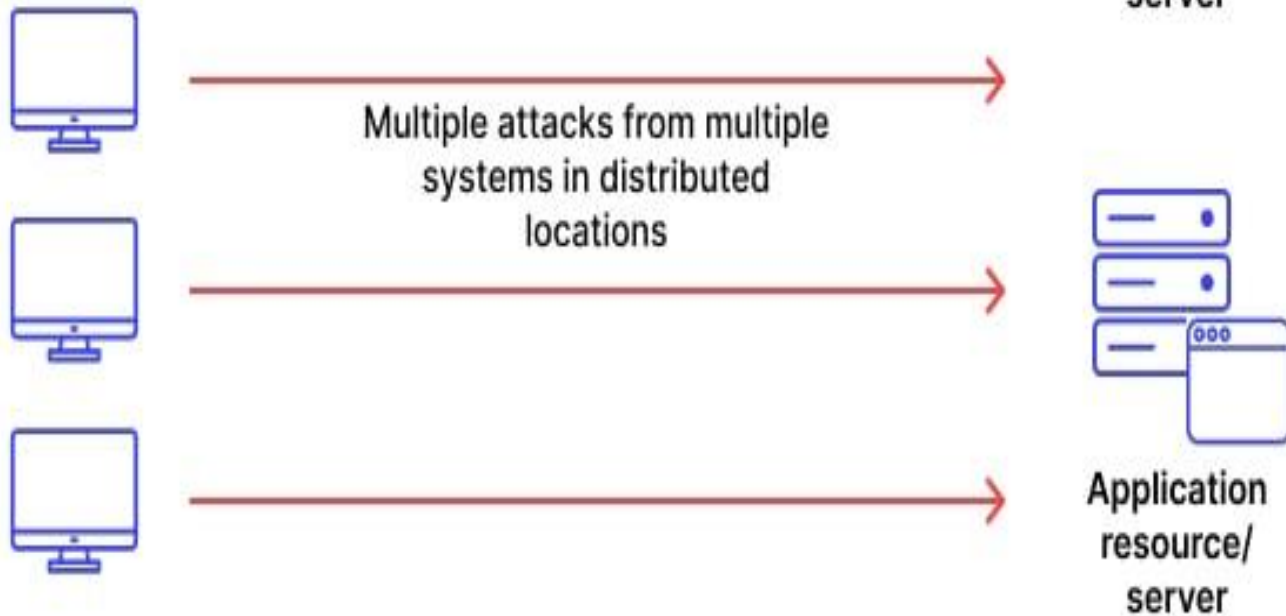
- **HTTP Flooding**
- **SYN Flooding**
- **UDP Flooding**
- **DNS Flooding**
- **Application Layer Attacks**
 - Ex. A **web application firewall (WAF)** protecting an online banking site is targeted.
 - Attackers send **complex HTTP requests** that exploit a specific **vulnerability** in the **WAF**.

DoS Vs. DDoS

DoS attack



DDoS attack



DoS Vs. DDoS

Stands for Denial of Service

A single machine is used
to launch an attack

Comparatively less
complicated

There is no malware
involvement

Stands for Distributed
Denial of Service

Multiple machines are used
to launch an attack

More complicated and
difficult to prevent

Uses malware to affect
multiple machines

Questions

Q1: Define cybersecurity?

Q2: Define the term “**confidential data**”?

Name some examples.

Q3: How could the hackers breach integrity of data?

Q4: List the main differences between DoS and DDoS?

Answers

A1:

- “The **protection of software, hardware**, and data **resources connected and stored** on the **Internet** is **known** as the **cybersecurity**”.
- The **protection** of the **personal, financial** data, **commercial** data, business-critical information, **operational** continuity, data integrity, and availability of **online software** services fall in the **cybersecurity domain**.

Answers

A2: Confidential data are **sensitive data** that should not be **known** by **someone** who is not **authorized** to **read** or **see that**. **Some examples** of confidential data include the following:

- **Personal identity information**
- **Credit card information**
- **Bank account information**
- **Personal health information**
- **Business or trade secrets**

Answers

A3: The **hackers achieve** the **data integrity** breach through **multiple ways** as mentioned below:

- **Installing malware** on the **server**
- **Malicious encryption** of data
- **Manipulation** of **original** data
- **Installing viruses** on the **hosts**
- **Insiders** with **malicious objectives**

Answers

A4:

Stands for Denial of Service

A single machine is used
to launch an attack

Comparatively less
complicated

There is no malware
involvement

Stands for Distributed
Denial of Service

Multiple machines are used
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More complicated and
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thanks for
listening!